

Calvin Nel

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Senior Technical Lead | Software Architect | .NET & Distributed Systems

Cape Town, South Africa

South African Citizen | British Citizen

Professional Summary

Senior Technical Lead and Software Architect with extensive experience designing, evolving and delivering complex software systems across fintech and high-volume transaction platforms.

My experience spans both legacy and modern .NET ecosystems, giving me a pragmatic understanding of how systems evolve in the real world. I specialise in software architecture, distributed systems, API design, system integration and technical leadership, with a focus on creating solutions that are reliable, maintainable and appropriate to the problem.

I bring a hands-on engineering perspective to architecture. Rather than applying patterns or technologies for their own sake, I focus on understanding the constraints of the existing system, the business requirements and the desired future state, then designing an architecture that can realistically get there.

I have led technical direction across complex platforms, balancing architectural improvements with backward compatibility, operational reliability, performance and the practical realities of evolving established systems.

Career Snapshot

- 20+ years in IT
 - 15+ years in professional software development
 - Technical Lead since 2022
 - Extensive fintech and payment platform experience
 - Backend architecture and distributed systems
 - 200+ production deployments orchestrated at a single fintech platform
 - South African & British Citizen
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Career History

Company	Position	Period
PayFast	Senior Lead Developer	Aug 2022 – Present
24.com (Media24)	Senior Software Developer	Oct 2019 – Jul 2022
OrderIn	Senior Software Developer	Jan 2019 – Sep 2019
CSG International	Senior Software Developer	Jun 2017 – Dec 2018
RCS	Senior Software Developer	Dec 2015 – May 2017
Alacrity	Analyst Developer	Mar 2011 – Nov 2015
Khusela Solutions	Intermediate Support Technician	Apr 2010 – Mar 2011
Parker Hannifin (UK)	Intermediate IT Technician	Apr 2009 – Nov 2009
Storeys:SSP (UK)	IT Technician	Aug 2004 – Apr 2009

Core Competencies

- Technical Leadership
 - Software Architecture
 - Backend Engineering
 - Distributed Systems
 - API Design & Integration
 - Payment Platforms
 - Legacy System Modernisation
 - Performance Optimisation
 - Technical Debt Reduction
 - DevOps Collaboration
 - Agile Delivery
 - Mentoring & Coaching
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Technical Skills

Frameworks

- .NET Framework 4.0–4.8
- .NET 5–10
- ASP.NET MVC 3-5
- ASP.NET Core MVC 5-10 + preview 11
- Entity Framework (EF) 4–6
- EF Core 5–9
- SignalR

Databases

- MS SQL Server
- MySQL / MariaDB
- Oracle
- Redis (distributed caching)

Frontend

- Vue.js 2/3
- JavaScript
- Angular
- HTML / CSS
- Legacy: jQuery, Knockout.js

Cloud & DevOps

- Azure
- Azure DevOps
- Docker
- Kubernetes (Rancher)
- GitHub
- CI/CD
- AWS

Architecture & Design

- REST APIs
 - Distributed Systems
 - Domain Driven Design (DDD)
 - Microservices
 - Event-Driven Architecture
 - SOLID Principles
 - Object-Oriented Design
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Professional Experience

Senior Lead Developer

PayFast

August 2022 – Present

Overview Technical Lead responsible for PayFast's EFT platforms, including SID, USSD and the wider EFT ecosystem.

Joined as a Senior Developer and immediately assumed Team Lead responsibilities following the departure of the existing lead. Since then, I have been responsible for technical direction, solution architecture, engineering standards and successful delivery across multiple payment products.

Responsibilities

- Lead the EFT engineering team.
- Drive technical architecture and solution design.
- Own sprint planning and technical delivery.
- Perform code reviews and maintain engineering standards.
- Mentor developers and support technical growth.
- Manage production deployments.
- Prepare Change Requests (CR) and Deployment Requests (DR).
- Collaborate with product owners, business stakeholders and engineering teams.

Key Achievements

- Led delivery of the Capitec Pay integration.
- Delivered Nedbank Direct alongside multiple concurrent initiatives.
- Delivered the Absa Pay integration.
- Drove rigorous code review and enforced testing discipline on all changes, removing the need for a dedicated QA function and delivering 200+ production deployments with virtually zero rollbacks and no downtime.
- Preserved backward compatibility of API contracts across the service dependency chain, enabling independent, zero-downtime deployments rather than coordinated lockstep releases.
- Resolved a critical SignalR information leakage issue.
- Improved performance across multiple high-volume payment workflows.
- Led architecture across multiple overlapping EFT systems.
- Adopted AI-assisted development tooling (Copilot, Cursor, Kiro) to reduce boilerplate and accelerate delivery, while maintaining rigorous human review so AI augments rather than replaces engineering judgement.

Technologies

- .NET Framework 4.7.2
 - .NET Core / .NET 5-8
 - SignalR
 - SQL Server
 - Redis
 - Vue.js 2
 - Kubernetes (Rancher)
 - REST APIs
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Senior Software Developer

24.com (Media24)

October 2019 – July 2022

Highlights

- Member of the Core Engineering team.
 - Modernised legacy applications into .NET Core APIs.
 - Applied Domain Driven Design (DDD) principles and contributed to a microservices architecture.
 - Developed and maintained backend services supporting high-traffic digital platforms.
 - Worked with Azure cloud services and Microsoft Azure DevOps for source control, work management and CI/CD.
 - Collaborated with cross-functional teams using Agile delivery practices.
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Senior Software Developer

OrderIn

January 2019 – September 2019

Highlights

- Worked across React, .NET MVC and ASP.NET Core services.
 - Assisted migration towards microservices architecture.
 - Implemented IdentityServer4 authentication and authorisation.
 - Worked with AWS-hosted containerised services.
 - Developed backend APIs supporting multiple applications.
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Senior Software Development Engineer

CSG International

June 2017 – December 2018

Highlights

- Developed enterprise applications using C#, VB.NET, SQL Server and Oracle.
- Optimised application and database performance.
- Performed code reviews and mentored developers.
- Shared engineering knowledge and best practices.

- Collaborated with international engineering teams.
 - Delivered GDPR-related platform improvements.
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Senior Database / Software Developer

RCS

December 2015 – May 2017

Highlights

- Sole developer responsible for major financial integration projects.
 - Designed and built the Smart Credit integration platform.
 - Delivered the Cash Advance platform.
 - Owned the full software lifecycle from architecture through production support.
 - Worked closely with retail and financial partners.
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Analyst Developer

Alacrity

March 2011 – November 2015

Highlights

- Delivered enterprise software for clients including Old Mutual, Bridgestone, I&J, IDS and WyzeTalk.
 - Worked directly with customers on solution design.
 - Participated in planning, architecture and implementation.
 - Developed full-stack solutions across the Microsoft technology stack.
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Education

Foundation Degree

ICT System Support

University of Northumbria

United Kingdom

Professional Development

Completed courses including:

- Software Architecture & Security
 - ASP.NET Core API Development
 - Blazor Development
 - RabbitMQ Microservices
 - Design Patterns in C#
 - Software Architecture Fundamentals
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Engineering Philosophy

- Build maintainable software before clever software.
 - Reduce technical debt continuously.
 - Optimise systems using evidence rather than assumptions.
 - Design systems that are easy to evolve.
 - Mentor engineers and improve team capability.
 - Automate repetitive work wherever practical.
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Professional Interests

I enjoy keeping up with developments across the .NET ecosystem and regularly explore new technologies and engineering practices to improve software quality and developer productivity.

Current areas of interest include:

- AI-assisted development tooling (Cursor, GitHub Copilot, Kiro)
 - Native AOT
 - Distributed systems
 - Performance optimisation
 - Modern API design
 - Open-source libraries
 - Benchmarking and profiling
 - Containerisation
 - Cloud-native development
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Contact

Calvin_Nel@msn.com

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